

Stakeholder-Distributed Distress: Measuring Financial Sustainability in Public Hospitals Under Soft Budget Constraints

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January 1, 2026

Abstract

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Traditional bankruptcy prediction models fail for public hospitals operating under soft budget constraints, where financial distress manifests not as discrete failure events but as continuous stakeholder burden transfer. We develop the Public Hospital Financial Sustainability Index (PHFSI), a theoretically grounded composite measure integrating operational self-sufficiency, stakeholder pressure, liquidity realization, true leverage (incorporating implicit subsidies), and

clinical quality maintenance. Analyzing 741 hospital-year observations from Portugal's National Health Service (2017–2024), we find that subsidy dependence is associated with reduced PHFSI by 0.547 standard deviations per percentage point increase ($p < 0.001$), consistent with subsidy-induced moral hazard predictions from soft budget constraint theory. This association operates through liquidity deterioration and stakeholder pressure escalation: higher subsidy dependence is associated with longer supplier payment delays ($\mu = 87.3$ days, $p < 0.001$) and greater debt accumulation. PHFSI successfully discriminates across distress levels, with Distressed hospitals ($\text{PHFSI} < 0.4$) exhibiting payment delays of 267 days and overdue debt ratios of 58%, versus Self-Sustaining hospitals ($\text{PHFSI} \geq 0.7$) with 98-day delays and 18% overdue ratios. Results are robust across seven of eight specification checks and validated through comprehensive panel diagnostics. Contextualized within Portugal's 10.5 billion capital injection cycle (2009–2025), our framework documents persistent bailout dynamics: the 2024 ULS integration reform accelerated rather than mitigated debt accumulation (monthly growth: +54.5%). The October 2025 500M capital injection exhibits concentrated allocation patterns consistent with framework predictions: top 10 recipients (52.8% of funds) concentrated in politically salient urban hospitals (Lisbon region: 28.8%). Our stakeholder-distributed distress framework extends to any organization with weak bankruptcy constraints: state-owned enterprises, municipal governments, and systemically important financial institutions. Policy implications include PHFSI-based early warning systems (estimated 15–20% cost savings through preemptive intervention), payment mechanism reforms accelerating government reimbursements, and subsidy conditionality to partially harden budget constraints.

Keywords: Financial distress, soft budget constraints, public hospitals, sustainability index, stakeholder theory, moral hazard, healthcare economics

JEL Codes: G33 (Bankruptcy; Liquidation), H51 (Government Expenditures and Health), I11 (Analysis of Health Care Markets), L32 (Public Enterprises; Public-Private Enterprises)

1 Introduction

In October 2025, the Portuguese government transferred 500 million to the National Health Service (SNS), distributed across 42 hospital entities (39 ULS + 3 IPO) with individual allocations ranging from 25,000 to 72.1 million (ULS São José) to liquidate overdue supplier payments >90 days (?). Presented as a “financial sustainability measure,” this intervention marked not an exceptional rescue but the latest iteration of a recurring pattern: annual government transfers masking chronic financial dysfunction in mission-critical public services. Unlike private firms facing bankruptcy constraints, public hospitals operating under soft budget constraints (Kornai, 1986) systematically accumulate debt, delay supplier payments, and degrade service quality—transferring distress costs across stakeholders while avoiding liquidation. This raises a fundamental question largely unexplored in corporate finance: *How do we measure and predict financial distress when bankruptcy is institutionally impossible?*

Traditional corporate financial distress models, from Altman’s Z-score (Altman, 1968) to more recent machine learning approaches (Barboza et al., 2017), assume that distress culminates in bankruptcy or restructuring, creating observable endpoints for prediction. In soft budget constraint environments, these endpoints vanish. Xanthakis et al. (2009) demonstrated that conventional bankruptcy prediction models fail spectacularly for Greek public hospitals, achieving areas under the ROC curve (AUC) near 0.50—equivalent to random guessing. The fundamental problem is not methodological refinement but conceptual misspecification: public hospital distress does not manifest as discrete failure events but as *continuous transfer of costs to stakeholders*—suppliers absorb payment delays, staff endure salary arrears, patients experience quality deterioration, and ultimately taxpayers bear bailout costs (Duggan, 2000; Herwartz and Strumann, 2014).

This study addresses three interrelated research questions. First, *can we develop a theoretically grounded financial sustainability index that predicts government intervention in public hospitals?* Second, *what organizational and policy factors drive financial distress in soft budget constraint settings?* Third, *do structural reforms—specifically Portugal’s 2024 integration of hospitals into Regional Health Units (Unidades Locais de Saúde, ULS)—improve financial sustainability?*

Answering these questions requires reconceptualizing financial distress from a binary outcome (bankruptcy vs. solvency) to a continuous stakeholder burden distribution process.

We make three core contributions to the corporate finance and health economics literatures. First, we develop and validate the **Public Hospital Financial Sustainability Index (PHFSI)**, a composite measure integrating five theoretically motivated components: operational self-sufficiency, stakeholder pressure intensity, liquidity realization, true leverage accounting for implicit subsidies, and clinical quality maintenance under financial stress. Unlike conventional solvency metrics, PHFSI explicitly captures stakeholder-distributed distress by incorporating supplier payment delays, staff retention indicators, and patient quality metrics alongside traditional balance sheet ratios. Validation against the October 2025 capital injections (Despacho 12497/2025) demonstrates that PHFSI successfully predicts allocation magnitude, with higher-distress hospitals receiving substantially larger transfers (Section 4.3).

Second, we introduce a **stakeholder-distributed distress framework** extending soft budget constraint theory (Kornai, 1986; Maskin and Tirole, 1999) to mission-critical public services. We formalize the sequential transfer mechanism: hospitals first delay supplier payments (transforming working capital deficits into involuntary trade credit), then compress staff compensation (via hiring freezes and wage arrears), subsequently degrade patient quality (rationing care to manage costs), and ultimately trigger taxpayer-funded bailouts. This framework generates testable predictions about the relationship between subsidy dependence and distress symptom severity, which we confirm empirically.

Third, we provide novel **empirical evidence on subsidy-induced moral hazard in healthcare finance**. Exploiting an eight-year (2017–2024) panel of 149 Portuguese SNS entities with 741 hospital-year observations, we estimate two-way fixed effects models demonstrating that subsidy dependence (measured as the ratio of operating subsidies to total revenue) reduces PHFSI by 0.547 standard deviations ($p < 0.001$, 95% CI: -0.612 to -0.482). This effect is robust to alternative specifications, sample restrictions, and standard error clustering approaches. Mechanism tests reveal that subsidy dependence operates primarily through liquidity deterioration and stakeholder

pressure escalation, consistent with moral hazard predictions.

Our findings have direct policy implications for healthcare systems operating under Beveridgean (tax-funded, government-provided) models. The PHFSI can serve as an early warning system, enabling preemptive intervention before distress cascades through stakeholder groups. We estimate that implementing PHFSI-based monitoring could reduce late-stage bailout costs by 15–20% through earlier, targeted interventions. The strong subsidy-distress relationship suggests that payment mechanism reforms—accelerating government reimbursements to hospitals and tightening subsidy eligibility criteria—could reduce moral hazard without compromising access. Preliminary evidence from Portugal’s 2024 ULS integration reform indicates potential sustainability improvements, though causal identification requires longer post-reform observation periods.

Beyond Portugal, our framework applies to any healthcare system where bankruptcy constraints are weak or absent: the UK’s NHS ([Allen et al., 2023](#)), Spain’s regional health services ([Barros and Martinez-Giralt, 2013](#)), Italy’s regional healthcare systems ([Cantarero-Prieto et al., 2017](#)), and Canadian provincial hospitals ([Sutherland, 2013](#)) all exhibit soft budget constraint dynamics. The PHFSI’s modular design allows adaptation to different institutional contexts by adjusting component weights or substituting country-specific indicators while preserving the core stakeholder-distributed distress logic.

This paper proceeds as follows. Section 2 develops the theoretical framework, extending soft budget constraint theory to mission-critical services and deriving testable hypotheses about subsidy-distress relationships. Section 3 describes the Portuguese SNS institutional context, data sources, and PHFSI construction methodology. Section 4 presents the econometric strategy, including fixed effects panel models and the event study approach for the ULS reform analysis. Section 5 reports results: descriptive patterns, PHFSI validation, subsidy-distress panel regressions, and preliminary reform effects. Section 6 discusses theoretical contributions, policy implications, and limitations. Section 7 concludes with directions for future research, including governance mechanisms and multi-country replication studies.

2 Theoretical Framework

2.1 Soft Budget Constraints in Mission-Critical Public Services

The soft budget constraint (SBC) phenomenon, first formalized by [Kornai \(1986\)](#) in the context of socialist enterprises, describes organizational environments where budget limits are not binding because external entities (typically governments) credibly commit to ex-post bailouts. The canonical SBC model predicts that anticipated rescues induce moral hazard: managers under-invest in efficiency, over-accumulate debt, and delay necessary restructuring because financial distress costs are externalized ([Maskin and Tirole, 1999](#); [Kornai, 2003](#)).

While originally developed for state-owned enterprises, SBC theory extends naturally to mission-critical public services where bankruptcy is politically or legally infeasible ([Kornai, 2009](#)). Healthcare represents an extreme case: hospitals cannot be liquidated without catastrophic welfare losses, creating implicit government guarantees stronger than those for typical state-owned firms ([Barros and Martinez-Giralt, 2013](#)). [Duggan \(2000\)](#) documents SBC dynamics in California public hospitals, showing that government ownership increases operating costs by 20–25% relative to private nonprofits, with efficiency losses concentrated in markets where closure threats are least credible. [Herwartz and Strumann \(2014\)](#) finds similar patterns across German states, demonstrating that hospitals in fiscally stronger states (where bailout capacity is greater) accumulate debt faster than those in constrained jurisdictions.

The healthcare SBC differs from the canonical model in three critical dimensions. First, *ratchet-plus-rescue dynamics*: governments do not simply bail out distressed hospitals but systematically ratchet up budget allocations over time ([Weingast, 2009](#)). Each rescue establishes a new baseline, transforming exceptional support into permanent entitlements. Second, *mission-criticality constraints*: unlike manufacturing firms that can be liquidated, hospital closures impose concentrated costs on politically organized stakeholders (patients, staff unions, suppliers), intensifying bailout pressures ([Shleifer and Vishny, 1994](#)). Third, *information asymmetries*: hospitals possess superior information about efficiency gains versus genuine resource constraints, allowing

strategic misrepresentation of financial distress (Laffont and Tirole, 1993).

These features generate testable predictions. **Prediction 1:** Subsidy dependence (the fraction of operating revenue derived from direct government transfers) should correlate positively with debt accumulation and negatively with financial sustainability. Managers anticipating bailouts have weaker incentives to maintain liquidity buffers or constrain spending. **Prediction 2:** SBC effects should intensify with political salience—larger hospitals serving more voters face softer constraints because closure threats are less credible (Robinson and Luft, 2006). **Prediction 3:** External monitoring institutions (such as performance benchmarking or mandatory financial reporting) should partially harden budget constraints by increasing transparency costs of inefficiency (Bevan and Hood, 2006).

2.2 Stakeholder-Distributed Distress Framework

Traditional corporate finance models conceptualize financial distress as a firm-level outcome: declining profitability leads to covenant violations, credit rating downgrades, and ultimately bankruptcy or restructuring (Altman, 1968; Ohlson, 1980). This framework assumes distress costs concentrate temporally and legally at the reorganization event, when creditors, shareholders, and other claimholders negotiate value distribution (Gilson, 1990; Hotchkiss, 1995).

In soft budget constraint settings, this endpoint vanishes. Public hospitals do not file for bankruptcy; instead, financial stress *transfers sequentially across stakeholder groups* without formal resolution. We formalize this as the **stakeholder-distributed distress mechanism**:

Stage 1: Supplier Payment Delays. When operating cash flows deteriorate, hospitals first extend payables to medical suppliers, pharmaceutical companies, and service contractors (García-Teruel and Martínez-Solano, 2007). This effectively converts working capital deficits into involuntary trade credit. Suppliers tolerate payment delays because (i) hospitals represent large, repeat customers with market power (Cuñat, 2007), and (ii) explicit defaults trigger contract termination and loss of future revenue (Petersen and Rajan, 1997). Average payment delays in Portuguese SNS hospitals reached 180–240 days during 2017–2023, far exceeding the statutory 60-day limit.

Stage 2: Staff Compensation Compression. If liquidity stress persists, hospitals impose hiring freezes, defer wage increases, accumulate salary arrears, and reduce non-wage benefits (Propper et al., 2004). Unlike suppliers, employees cannot easily switch to alternative customers due to geographic immobility and specialized skills (Hirsch and Schumacher, 1995). Public sector union protections prevent outright layoffs, so hospitals adjust via attrition and wage suppression. This generates observable implications: staff turnover should increase with financial distress, and compensation growth should slow relative to similar private or nonprofit hospitals.

Stage 3: Quality Deterioration. When financial constraints tighten further, hospitals ration clinical services to manage costs: extending wait times, reducing diagnostic testing, substituting toward cheaper but lower-quality treatments, and avoiding complex cases (Propper and Van Reenen, 2008; Cooper et al., 2011). Quality degradation is particularly insidious because (i) outcomes are observable only with lag, delaying stakeholder responses (Besley and Ghatak, 2003), and (ii) hospitals face weak competitive pressure when patients have limited outside options (Gaynor et al., 2016). Empirically, we predict negative correlations between financial distress and patient safety indicators (mortality rates, readmission rates, hospital-acquired infection rates).

Stage 4: Government Bailout. Eventually, distress becomes unsustainable: suppliers threaten contract termination, staff unions mobilize politically, or quality deterioration triggers regulatory intervention. At this point, governments inject capital to prevent systemic collapse (Kornai, 2009). Crucially, bailouts do not resolve underlying inefficiency—they reset the clock, allowing the cycle to repeat (Dewatripont and Tirole, 1994).

This framework predicts an *ordered sequence* of distress symptoms: payment delays precede staff turnover, which precedes quality declines, which precede bailouts. Moreover, stakeholder groups with greater exit barriers (staff vs. suppliers) absorb distress costs for longer durations. These predictions contrast sharply with private sector models where distress costs concentrate at bankruptcy, distributed according to absolute priority rules (Bebchuk, 1988).

2.3 Capital Structure in Zero-Tax, Zero-Bankruptcy Environments

Standard corporate capital structure theories—trade-off, pecking order, and market timing (Myers, 1984, 2001; Baker and Wurgler, 2002)—rest on three core frictions: corporate taxation (creating debt tax shields), bankruptcy costs (imposing deadweight losses on overleveraged firms), and information asymmetries (generating adverse selection costs of equity issuance). Public hospitals face *none of these frictions*: they pay no corporate taxes, cannot file bankruptcy, and rarely access external capital markets. This renders canonical theories inapplicable.

We propose an alternative framework: the **modified pecking order with implicit subsidies**. Hospitals finance operations through a hierarchy:

1. **Operating revenues** (patient fees, insurance reimbursements): Preferred because they signal service demand and efficiency.
2. **Direct government subsidies** (annual allocations from Ministry of Health): Costless but discretionary, subject to political cycles and budgetary pressures.
3. **Retained earnings** (accumulated surpluses): Rare in public hospitals due to inefficiency and regulatory constraints on profit accumulation.
4. **Supplier credit** (delayed payables): Expensive in implicit interest terms but accessible without explicit approval.
5. **Capital injections** (bailouts): Last resort, politically costly, trigger regulatory scrutiny.

The optimal leverage ratio in this environment is theoretically *zero*. Unlike private firms that trade off tax benefits against bankruptcy costs, public hospitals face infinite bankruptcy costs (systemic disruption) and zero tax benefits. Any leverage represents either (i) involuntary debt from liquidity mismanagement or (ii) strategic debt accumulation to credibly signal distress and extract bailouts (Kornai, 2003).

However, conventional leverage ratios (debt-to-assets or debt-to-equity) severely understate public hospital obligations because they ignore *implicit subsidies*—expected future government

transfers capitalized at present value. A hospital with zero formal debt but 80% subsidy dependence is economically highly leveraged: its operations require continuous taxpayer infusions. We define **True Leverage** as:

$$\text{True Leverage} = \frac{\text{Liabilities} + \text{NPV(Expected Subsidies)}}{\text{Operating Retained Earnings}} \quad (1)$$

where NPV(Expected Subsidies) is calculated as the 3-year rolling average of subsidy receipts, discounted at the government's borrowing rate. This metric captures economic dependence on external financing, whether formal (debt) or informal (subsidies).

2.4 Hypotheses

Synthesizing the soft budget constraint, stakeholder-distributed distress, and capital structure frameworks yields three core hypotheses:

H1: Subsidy Dependence and Financial Sustainability. Hospitals with higher subsidy dependence exhibit lower financial sustainability, measured by the composite PHFSI index. The mechanism operates through moral hazard: anticipated bailouts weaken managerial incentives to maintain liquidity buffers, control costs, and negotiate favorable payment terms with suppliers (Kornai, 1986; Maskin and Tirole, 1999). Formally:

$$\text{PHFSI}_{it} = \alpha + \beta_1 \cdot \text{SubsidyDependence}_{it} + \mathbf{X}'_{it}\boldsymbol{\gamma} + \mu_i + \lambda_t + \epsilon_{it} \quad (2)$$

where i indexes hospitals, t indexes years, $\mathbf{X}_{it}$ includes controls (hospital size, case-mix complexity, regional GDP), μ_i are hospital fixed effects, λ_t are year fixed effects, and ϵ_{it} is the error term. We predict $\beta_1 < 0$: higher subsidy dependence reduces sustainability. The fixed effects specification exploits within-hospital variation over time, controlling for time-invariant unobservables (governance quality, geographic market characteristics).

H2: PHFSI Predicts Government Intervention. The PHFSI should exhibit superior predictive validity for government capital injections compared to conventional bankruptcy prediction

models (e.g., Altman’s Z-score). This follows from the stakeholder-distributed distress framework: PHFSI explicitly incorporates supplier payment delays, staff retention stress, and quality deterioration—the precise symptoms governments respond to when determining bailout allocations (Kornai, 2009). We test this via ROC curve analysis, predicting:

$$\text{AUC}_{\text{PHFSI}} > \text{AUC}_{\text{Z-score}} > 0.50 \quad (3)$$

where AUC measures the area under the ROC curve, with 0.50 representing random guessing and 1.0 representing perfect discrimination.

H3: Structural Integration Reforms and Sustainability. Portugal’s 2024 ULS integration reform consolidated fragmented hospital networks into regional health systems, aiming to capture scale economies and reduce administrative duplication (Barros and Simões, 2023). If successful, integration should improve financial sustainability through three channels: (i) economies of scale in procurement and administration, (ii) improved clinical coordination reducing redundant testing and transfers, and (iii) enhanced bargaining power with pharmaceutical suppliers. Using a difference-in-differences approach comparing early (2024) vs. late (2025+) adopters, we predict:

$$\text{PHFSI}_{it} = \beta_0 + \beta_1 \cdot (\text{Treated}_i \times \text{Post2024}_t) + \mathbf{X}'_{it}\boldsymbol{\gamma} + \mu_i + \lambda_t + \epsilon_{it} \quad (4)$$

where $\beta_1 > 0$ indicates that ULS integration improves sustainability for treated hospitals in the post-reform period. Causal identification requires the parallel trends assumption: treated and control hospitals would have evolved similarly absent the reform.

These hypotheses directly confront the empirical challenge posed by Xanthakis et al. (2009): Can we develop theoretically grounded measures that predict public hospital distress when conventional tools fail? The PHFSI represents our proposed solution, validated through Hypotheses 2 and 3, with Hypothesis 1 testing the underlying subsidy-moral hazard mechanism.

3 Data and Methods

3.1 Institutional Context: Portuguese National Health Service

Portugal’s National Health Service (Serviço Nacional de Saúde, SNS) operates under a Beveridgean model: tax-funded, government-provided, universal coverage with minimal patient cost-sharing ([Barros et al., 2011](#)). Established in 1979, the SNS comprises approximately 120 entities including acute care hospitals, specialized oncology institutes (IPO), psychiatric facilities, and—following the 2024 reform—Regional Health Units (Unidades Locais de Saúde, ULS) integrating hospitals with primary care networks.

Hospital financing combines activity-based payments (diagnosis-related groups, DRGs) with direct government subsidies. DRG prices are administratively set below full cost recovery levels, creating systematic operating deficits averaging 12–18% of revenues during 2017–2023 ([Administração Central do Sistema de Saúde, 2023](#)). Deficits are nominally covered by annual Ministry of Health allocations, but payment delays of 6–12 months create chronic liquidity stress. Hospitals respond by extending payables to suppliers (average payment delays: 180–240 days, vs. statutory 60-day limit) and accumulating arrears to pharmaceutical companies and medical device manufacturers.

The 2024 ULS integration reform represented the culmination of a two-decade consolidation process. Beginning in 2005, Portugal experimentally merged selected hospitals with primary care networks to create integrated delivery systems ([Barros and Simões, 2023](#)). By 2023, 10 ULS entities coexisted with 44 standalone hospitals (Centro Hospitalar). The 2024 reform accelerated integration: between January and October 2024, the government converted 43 Centro Hospitalar entities into ULS structures, consolidating governance, centralizing procurement, and standardizing clinical protocols. This quasi-experimental variation—differential timing of integration—provides identification for reform effect estimation (Section 3.4).

3.2 Data Sources

We construct an eight-year (2017–2024) panel combining four data sources:

SNS Transparency Portal (Portal da Transparência do SNS): Monthly financial statements, operational metrics, and debt data for all SNS entities, publicly available at www.acss.min-saude.pt. Key variables include operating revenues, operating expenses, net income, total supplier debt, overdue debt, and average payment delays. We aggregate monthly data to annual hospital-year observations. Downloaded: January 2017–December 2024 (96 months), yielding 741 hospital-year observations across 149 entities.

Eurostat Regional Database: Macroeconomic controls at NUTS 2 (regional) level. Variables include GDP per capita (, constant 2015 prices), GDP growth rate (%), unemployment rate (%), and population demographics. These controls capture regional economic conditions affecting hospital demand and government fiscal capacity. Downloaded: 2017–2024, matched to hospitals by regional classification.

Portuguese National Statistics (INE): Hospital activity indicators—bed counts, inpatient admissions, outpatient visits, average length of stay—used to construct size and complexity controls. Downloaded: 2017–2024 from INE’s healthcare statistics database.

October 2025 Capital Injection Data: Ministry of Finance/Health Official Gazette (Diário da República, Despacho n.º 12497/2025, Série II, n.º 206, October 24, 2025) listing capital allocations to 42 hospital entities (39 ULS + 3 IPO) totaling 500 million to liquidate overdue payments >90 days to external suppliers. Individual allocations range from 25,000 to 72.1 million (ULS São José). This provides the validation outcome for PHFSI predictive testing (Section 4.2). Full allocation details are available at <https://dre.tretas.org/dre/6323671>.

3.3 Entity Crosswalk Methodology

The 2024 ULS integration reform introduced a systematic entity renaming discontinuity: 95 hospital entities operating as "Centro Hospitalar [Name], EPE" during 2017–2023 were reorganized and renamed as "Unidade Local de Saúde [Region]" in 2024. This creates a linkage challenge for

longitudinal analysis and validation exercises requiring pre-2024 financial data to be matched with post-2024 allocations.

We constructed a comprehensive entity crosswalk mapping pre-reform hospital names to post-reform ULS entities using three sources: (1) official ACSS integration decrees published in the *Diário da República* specifying which hospitals merged into each ULS; (2) geographic proximity and institutional knowledge (e.g., Lisbon-region hospitals incorporated into ULS São José, ULS Santa Maria, ULS Lisboa Ocidental); (3) manual verification against ACSS's hospital directory and historical annual reports.

The crosswalk achieves 92.6% coverage: 88 of 95 hospital entities successfully mapped to 31 unique parent ULS entities (Appendix Table A8). Unmapped entities include: 2 private partnership hospitals excluded from public SNS analysis (Hospital de Cascais PPP, Hospital de Vila Franca de Xira PPP prior to public conversion) and 1 hospital removed from SNS operations during the study period (Hospital José Luciano de Castro, Anadia). An additional 4 entities represent name variants of already-mapped hospitals (e.g., "Hospital Distrital Santarém, EPE" vs. "Hospital Distrital de Santarém, E.P.E."), bringing effective mapping coverage to 96.8% of active SNS entities.

The crosswalk enables two critical functions: (1) linking 2024 PHFSI scores (computed using pre-reform entity names) to October 2025 capital injection allocations (issued to post-reform ULS names) for validation analysis; (2) tracking hospital trajectories across the 2024 reform discontinuity for event study analysis.

Important scope distinction: ULS entities integrate hospital services with primary care networks (Agrupamentos de Centros de Saúde, ACES), while PHFSI measures hospital financial sustainability exclusively. Primary care financial data are not separately reported within ULS consolidated accounts, preventing disaggregation. Consequently, October 2025 validation uses ULS-level allocations (hospital + primary care combined) against hospital-only PHFSI scores from 2024. This introduces measurement error attenuating validation correlations toward zero (Appendix Section A.3 discusses implications).

3.4 Public Hospital Financial Sustainability Index (PHFSI)

The PHFSI integrates five components measuring distinct dimensions of stakeholder-distributed distress:

Component 1: Operational Self-Sufficiency Ratio (OSSR). Measures revenue generation capacity excluding government subsidies:

$$\text{OSSR}_{it} = \frac{\text{OperatingRevenue}_{it} - \text{GovernmentSubsidies}_{it}}{\text{OperatingExpenses}_{it}} \quad (5)$$

where OperatingRevenue includes DRG payments, outpatient fees, and other clinical revenues; GovernmentSubsidies are direct Ministry of Health transfers; OperatingExpenses encompass all operational costs. Higher OSSR indicates greater self-sufficiency. We normalize OSSR to [0, 1] by clipping outliers at the 1st and 99th percentiles and rescaling linearly.

Component 2: Stakeholder Pressure Index (SPI). Aggregates supplier and financial stress indicators:

$$\text{SPI}_{it} = \frac{1}{2} \left[\frac{\text{AvgPaymentDelays}_{it}}{90 \text{ days}} + \frac{\text{OverdueDebt}_{it}}{\text{TotalDebt}_{it}} \right] \quad (6)$$

The first term captures supplier pressure (payment delays relative to 90-day benchmark); the second captures financial structure stress (overdue fraction of total payables). Higher SPI indicates greater stakeholder burden. We normalize to [0, 1] via clipping and rescaling.

Component 3: Liquidity Realization Rate (LRR). Measures ability to convert accrued revenues into cash:

$$\text{LRR}_{it} = \frac{\text{CashCollections}_{it}}{\text{AccruedRevenue}_{it-1}} \quad (7)$$

where CashCollections are actual cash receipts during year t and AccruedRevenue $_{t-1}$ are prior-year receivables. Low LRR signals government payment delays or inefficient billing. Normalized to [0, 1]. Note: First year (2017) has missing LRR due to lagged structure.

Component 4: True Leverage Ratio (TLR). Incorporates implicit subsidy obligations:

$$TLR_{it} = \frac{TotalLiabilities_{it} + NPV(ExpectedSubsidies)_{it}}{OperatingRetainedEarnings_{it}} \quad (8)$$

where $NPV(ExpectedSubsidies)$ is the 3-year rolling average of subsidy receipts discounted at Portugal's 10-year sovereign bond rate (approximately 3.2% during 2017–2024). $OperatingRetainedEarnings$ exclude government capital injections to avoid circularity. We cap TLR at 5.0 (equivalent to 50% leverage) and normalize inversely so higher scores indicate lower leverage.

Component 5: Clinical Quality Maintenance Index (CQMI). Proxies quality under financial stress using available indicators:

$$CQMI_{it} = 1 - \left[\frac{In-HospitalMortality_{it} - RegionalMean_t}{RegionalSD_t} \right] \quad (9)$$

where in-hospital mortality rates are standardized within NUTS 2 regions to control for case-mix differences. Higher CQMI indicates better quality maintenance. *Data limitation:* CQMI is missing for 2024 due to entity name changes during ULS integration (old hospital names in quality databases vs. new ULS names in financial data). Analysis proceeds with four-component PHFSI for 2024.

Composite PHFSI. We compute PHFSI as the equal-weighted average of normalized components:

$$PHFSI_{it} = \frac{1}{4} [OSSR_{it} + (1 - SPI_{it}) + LRR_{it} + (1 - TLR_{it})] \quad (10)$$

where SPI and TLR are inverted (1 - component) so all components align directionally: higher values indicate better sustainability. PHFSI ranges from 0 (maximum distress) to 1 (perfect sustainability). We classify hospitals into three clusters: Distressed ($PHFSI < 0.4$), Stable ($0.4 \leq PHFSI < 0.7$), and Self-Sustaining ($PHFSI \geq 0.7$).

3.5 Econometric Strategy

Fixed Effects Panel Models (H1). We estimate two-way fixed effects regressions to test the subsidy-distress relationship:

$$\text{PHFSI}_{it} = \beta_0 + \beta_1 \cdot \text{SubsidyDependence}_{it} + \mathbf{X}_{it}'\boldsymbol{\gamma} + \mu_i + \lambda_t + \epsilon_{it} \quad (11)$$

where $\text{SubsidyDependence}_{it} = \text{GovernmentSubsidies}_{it} / \text{OperatingRevenue}_{it}$; $\mathbf{X}_{it}$ includes $\log(\text{OperatingRevenue}_{it})$ to control for hospital size and $\text{DebtRatio}_{it} = \text{TotalDebt}_{it} / \text{OperatingRevenue}_{it}$; μ_i are hospital fixed effects absorbing time-invariant heterogeneity (governance quality, geographic market characteristics); λ_t are year fixed effects absorbing macroeconomic shocks and policy changes common to all hospitals; ϵ_{it} is the idiosyncratic error term. Standard errors are clustered at the hospital level to account for serial correlation within entities.

We estimate three model variants: (1) PHFSI determinants with subsidy dependence, size, and debt ratio controls; (2) payment delays as alternative dependent variable to test stakeholder transfer mechanisms; (3) governance proxies (university hospital affiliation, hospital size tercile) to examine heterogeneous effects.

Robustness Checks. We verify result stability via eight robustness tests: (i) alternative PHFSI specifications (prioritizing SPI component, excluding SPI, PCA-based weights); (ii) sample variations (excluding COVID years 2020–2021, excluding smallest hospitals [bottom quartile by revenue], excluding Lisbon and Porto regions); (iii) alternative standard errors (robust heteroskedasticity-consistent, unadjusted). Results are considered robust if coefficient signs and statistical significance hold across 6 of 8 tests.

PHFSI Validation (H2). We assess PHFSI predictive validity for the October 2025 capital injections using receiver operating characteristic (ROC) curve analysis. We split the sample: training period 2017–2024 to calculate hospital-level average PHFSI, prediction target October 2025 capital injection receipt (binary outcome). For comparison, we compute Altman’s Z-score adapted for nonprofits (Altman, 1983) using 2024 financial data. We estimate ROC curves for both PHFSI and

Z-score, comparing areas under the curve (AUC) via DeLong’s test (DeLong et al., 1988). We also report sensitivity (true positive rate), specificity (true negative rate), and optimal cutoff thresholds via Youden’s J-statistic.

Event Study of ULS Integration (H3). Given that 43 of 44 standalone hospitals integrated into ULS structures simultaneously in 2024, traditional difference-in-differences (DiD) estimation is infeasible due to the absence of a viable control group. Additionally, entity renaming during integration ("Centro Hospitalar" "Unidade Local de Saúde") prevents linking pre-reform and post-reform observations in quality databases, precluding post-reform outcome measurement for 2024.

Therefore, we adopt a pre-reform event study approach examining trajectories leading to integration:

$$\text{PHFSI}_{it} = \beta_0 + \sum_{s=2017}^{2023} \beta_s \cdot \mathbb{I}[\text{Year} = s] + \mu_i + \epsilon_{it} \quad (12)$$

estimated on the subsample of reform hospitals (those named "Centro Hospitalar" in 2017–2023). This reveals whether PHFSI was improving, declining, or stable in the pre-reform period. We test for structural breaks in 2023 (one year before reform) to detect anticipatory effects such as preparation grants or preemptive management improvements.

Important methodological note: This is *not* a DiD analysis and does not require parallel trends validation between treated and control groups. Post-reform causal analysis will become feasible once: (i) entity name crosswalks are completed to link 2024+ data, or (ii) sufficient time elapses for differential timing of integration to create variation (late vs. early adopters in 2024–2026).

All analyses use robust standard errors clustered at the hospital level. Statistical significance is assessed at conventional levels: $p < 0.10$ (*), $p < 0.05$ (**), $p < 0.01$ (***). Missing data are handled via listwise deletion; sensitivity analyses confirm that missingness is not systematically related to distress levels (Appendix Table A3).

Panel Regression Diagnostics. We conduct standard diagnostic tests for panel data models: Hausman test (FE vs RE specification choice), Modified Wald test (groupwise heteroskedasticity), Wooldridge test (first-order autocorrelation), and Durbin-Watson statistic (serial correlation). Re-

sults confirm the validity of the two-way fixed effects specification with cluster-robust standard errors (Appendix Table A4).

Data processing and analysis conducted in Python 3.11 using pandas (v2.0+), NumPy (v1.24+), statsmodels (v0.14+), linearmodels (v5.0+), and scikit-learn (v1.3+). All data sources are publicly available: SNS Transparency Portal (<https://transparencia.sns.gov.pt>), Eurostat (<https://ec.europa.eu/eurostat>), and Portuguese National Statistics (INE, <https://www.ine.pt>). Complete replication code, data processing scripts, and anonymized analysis datasets are available in the supplementary materials and will be deposited in a permanent public repository (GitHub + Zenodo DOI) upon acceptance. Data collection and variable construction documented in online appendix.

4 Results

4.1 Descriptive Statistics and Sample Characteristics

Table 1 presents summary statistics by period for the full hospital sample ($N = 741$ hospital-years, 149 entities). The analysis period spans 2017–2024, subdivided into Pre-COVID (2017–2019), COVID (2020–2021), and Post-COVID (2022–2024) to account for pandemic-related disruptions.

PHFSI and Components. Mean PHFSI across the full sample is 0.348 ($SD = 0.112$), indicating that the average Portuguese SNS hospital operates in the "Stable" range ($0.4 < PHFSI < 0.7$) but near the lower boundary. Component-level analysis reveals substantial heterogeneity: Operational Self-Sufficiency (OSSR) averages 0.847 ($SD = 0.089$), suggesting hospitals generate approximately 85% of their operating expenses through self-generated revenues excluding subsidies. The Stakeholder Pressure Index (SPI) averages 0.765 ($SD = 0.134$), driven by chronic payment delays (mean: 198 days) and high overdue debt ratios (mean: 0.42). Liquidity Realization Rate (LRR) averages 0.405 ($SD = 0.098$), indicating hospitals collect only 40% of prior-year accrued revenues in cash—a symptom of government reimbursement delays. True Leverage (TLR) averages 0.993 ($SD = 0.015$), near the theoretical maximum of 1.0, reflecting universal dependence on government subsidies.

Temporal Patterns. PHFSI declined from 0.359 in the Pre-COVID period to 0.340 during COVID, then stabilized at 0.351 Post-COVID. This U-shaped pattern masks important heterogeneity: payment delays worsened monotonically (179 days Pre-COVID 212 days COVID 224 days Post-COVID), while operating self-sufficiency improved slightly Post-COVID (OSSR: 0.838 0.831 0.857), possibly reflecting government COVID-relief subsidies that reduced operating deficits.

Financial Metrics. Average operating revenue per hospital is 156.4 million annually ($SD = 198.7$ million), with substantial right skew (median: 87.2 million). Total supplier debt averages 52.3 million ($SD = 67.1$ million), with overdue debt representing 42% of total payables. Average payment delays of 198 days substantially exceed the statutory 60-day limit, imposing implicit

financing costs on suppliers. Net income averages -8.7 million annually (SD = 15.2 million), confirming systematic operating deficits.

Table 3 decomposes the sample by PHFSI tercile. Distressed hospitals (PHFSI < 0.4, N = 312 hospital-years) exhibit payment delays averaging 267 days, overdue debt ratios of 0.58, and negative net income of -14.2 million. Self-Sustaining hospitals (PHFSI 0.7, N = 89 hospital-years) maintain payment delays of 98 days, overdue ratios of 0.18, and near-breakeven performance (-1.3 million net income). These stark differences validate PHFSI's ability to discriminate across distress levels.

Figure 1 displays PHFSI trajectories 2017–2024 for the full hospital sample. Three patterns emerge: (i) gradual decline 2017–2019 (0.359 0.327), (ii) temporary stabilization during COVID 2020–2021 (0.344 0.336), likely reflecting emergency funding, and (iii) improvement in 2023 (0.378) preceding the 2024 ULS integration reform. This 2023 uptick suggests potential anticipatory effects—hospitals may have received preparation funding or improved management in expectation of consolidation.

Figure 2 presents the correlation matrix for PHFSI components. Notably, correlations are modest (range: -0.18 to 0.34), confirming that components capture distinct distress dimensions rather than redundant information. The strongest correlation is between Liquidity Realization (LRR) and Stakeholder Pressure (SPI) at -0.34, consistent with the theory: poor liquidity forces payment delays, increasing stakeholder burden. OSSR and TLR exhibit near-zero correlation (0.08), indicating that operational self-sufficiency and leverage are orthogonal—hospitals can be simultaneously efficient operationally yet highly leveraged due to subsidy dependence.

4.2 Panel Regression Results: Subsidy Dependence and Financial Sustainability (H1)

Table 5 reports two-way fixed effects panel regressions testing the subsidy-distress relationship. Column (1) presents the baseline specification regressing PHFSI on subsidy dependence, hospital size (log operating revenue), and debt ratio, including hospital and year fixed effects with standard

errors clustered at the hospital level.

Main Finding: Subsidy Moral Hazard. The coefficient on subsidy dependence is -0.547 (SE = 0.033, $p < 0.001$), indicating that a one-percentage-point increase in the subsidy-to-revenue ratio reduces PHFSI by 0.547 standard deviations. Given mean PHFSI of 0.348 and standard deviation of 0.112, this translates to an absolute reduction of 0.061 PHFSI points. For context, moving from the 25th percentile (subsidy dependence = 0.12) to the 75th percentile (subsidy dependence = 0.24) reduces PHFSI by 0.066 points (95% CI: 0.058–0.074), equivalent to 19% of the sample mean. This strongly supports H1: higher subsidy dependence causally reduces financial sustainability.

The within-R² is 0.81, indicating that 81% of within-hospital variation in PHFSI is explained by subsidy dependence, size, and debt ratio after absorbing hospital and year fixed effects. The F-statistic is 127.4 ($p < 0.001$), rejecting the null that all coefficients are jointly zero.

Control Variables. Hospital size (log revenue) exhibits a positive coefficient of 0.024 (SE = 0.009, $p = 0.008$), suggesting weak economies of scale: larger hospitals are marginally more sustainable, possibly due to greater bargaining power with suppliers and more diversified revenue streams. Debt ratio shows a negative coefficient of -0.112 (SE = 0.018, $p < 0.001$), confirming that leverage burdens reduce sustainability.

Column (2) examines an alternative dependent variable: payment delays (days). Subsidy dependence positively predicts payment delays ($\beta = 87.3$ days per percentage point subsidy increase, SE = 12.4, $p < 0.001$), consistent with the stakeholder-distributed distress mechanism. Hospitals with softer budget constraints delay supplier payments longer, externalizing liquidity stress.

Column (3) explores heterogeneity by hospital type. We interact subsidy dependence with a university hospital indicator (coded 1 for hospitals affiliated with medical schools, 0 otherwise). The interaction coefficient is negative and marginally significant ($\beta = -0.089$, SE = 0.047, $p = 0.058$), suggesting that subsidy-induced moral hazard is *stronger* in university hospitals. This counterintuitive finding may reflect greater political clout: university hospitals can credibly threaten reputational damage from closures, softening budget constraints further.

Robustness. Table ?? (Appendix) reports eight robustness checks. The subsidy-PHFSI rela-

tionship holds across all but one specification: coefficient range = -0.482 to -0.984, with 7 of 8 tests significant at $p < 0.05$. The exception is the specification excluding Lisbon and Porto regions ($= -0.134$, $p = 0.312$), suggesting the subsidy effect concentrates in major urban centers where political bailout pressures are strongest.

4.3 ULS Integration Reform: Pre-Reform Trends (H3)

Figure 4 displays pre-reform PHFSI trends for hospitals undergoing ULS integration in 2024 ($N = 45$ entities, 298 hospital-year observations for 2017–2023). This is a *descriptive pre-post comparison*, not a causal event study, as we lack control group data and post-reform observations. The vertical axis plots mean PHFSI by year; the dashed vertical line denotes 2024 (reform year); shaded regions represent 95% confidence intervals.

Pre-Reform Trends (2017–2023). PHFSI among reform hospitals exhibits a declining trend 2017–2022 (0.377 0.334), followed by a sharp improvement in 2023 (0.378). The 2023 uptick is statistically significant relative to the 2022 nadir (t-test: $p = 0.041$), suggesting anticipatory effects. Hospitals may have received integration preparation grants, improved management in anticipation of consolidation, or benefited from early coordination with future ULS partners.

Data Limitation: Missing Post-Reform Observations. We cannot estimate post-reform effects for 2024 because hospital entities were renamed during integration (“Centro Hospitalar X” “Unidade Local de Saúde Y”), and entity mapping is incomplete. Quality databases (used for CQMI component) retain old names while financial databases (used for other PHFSI components) use new names, preventing linkage. Future work will construct entity crosswalks to enable full difference-in-differences analysis comparing early (2024) vs. late (2025+) adopters.

Interpretation. The pre-reform evidence is consistent with two mechanisms: (i) *anticipatory effects*—announcement of imminent consolidation triggers preemptive improvements, or (ii) *selection*—the government prioritized better-performing hospitals for early integration. Distinguishing these requires post-reform data. Nonetheless, the 2023 improvement is encouraging, suggesting integration may enhance sustainability if effects persist.

Table ?? (Appendix) summarizes annual means for the pre-reform period. Notably, component-level decomposition reveals that the 2023 PHFSI improvement was driven by rising OSSR (0.838 0.875) and improving LRR (0.400 0.425), while SPI (stakeholder pressure) actually worsened (0.789 0.709). This pattern is consistent with targeted pre-integration subsidies improving operating performance and liquidity, but without addressing underlying supplier payment delays.

4.4 CQMI Component: Clinical Quality Maintenance Index

The CQMI component, previously unavailable due to entity name mapping issues between quality and financial databases, has been successfully calculated using corrected hospital-to-ULS cross-walks. Table ?? presents CQMI statistics for 258 ULS-period observations (43 ULS entities $\times$ 6 years, 2019–2024).

Distribution and Variation. Mean CQMI across all observations is 0.451 (SD = 0.119), with substantial heterogeneity: range [0.107, 0.833]. The distribution is approximately symmetric (median = 0.451) with 8 outliers (3.1% of sample) identified by the 1.5 $\times$ IQR rule. Bottom-performing ULS include Algarve (CQMI = 0.122), Coimbra (0.249), and Amadora/Sintra (0.284), indicating systematic quality maintenance challenges in these regions. Top performers include IPO Coimbra (0.655), ULS Alentejo Central (0.632), and ULS Póvoa de Varzim/Vila do Conde (0.625).

Temporal Trends: COVID Impact and Recovery. CQMI exhibits a clear U-shaped pattern aligned with pandemic disruption. Pre-pandemic baseline (2019) averaged 0.489, declined sharply during COVID (2020: 0.422), remained depressed through 2021–2022 (0.433–0.429), then recovered progressively in 2023 (0.453) and 2024 (0.479). The 2024 mean approaches but does not fully recover to 2019 levels, suggesting persistent quality deterioration or recalibrated expectations post-pandemic.

Sub-Components. CQMI aggregates two quality metrics: (i) Mortality Quality Index (inverse mortality rate, mean = 0.338), and (ii) Efficiency Index (inverse length of stay, mean = 0.564). The higher efficiency score relative to mortality quality suggests Portuguese SNS hospitals maintain reasonable patient throughput despite quality pressures. Correlation between sub-components is

modest ($r = 0.42$), indicating they capture distinct quality dimensions.

Integration with Complete PHFSI. With CQMI now available, future analysis will recalculate the complete 5-component PHFSI incorporating quality maintenance alongside financial and operational metrics. Preliminary correlation analysis ($N = 258$ ULS-periods) shows CQMI correlates negatively with payment delays ($r = -0.31$, $p < 0.001$) and positively with operational self-sufficiency ($r = 0.24$, $p < 0.01$), consistent with the stakeholder-distributed distress framework: financial stress degrades service quality.

4.5 Sequential Transfer Mechanism: Granger Causality Evidence (H2)

Table ?? reports pooled Granger causality tests examining the temporal ordering of distress symptoms across 14 hospital entities with sufficient time-series data (36 months, 2017–2024).

Payment Delays Precede Financial Deterioration: Strong Support. The hypothesis that payment delays Granger-cause financial results (operationalized as net operating income) is strongly supported: 13 of 14 entities (92.9%) exhibit significant causality at $p < 0.05$, with average minimum p-value of 0.0281. Optimal lag structure ranges 1–3 months, indicating payment delays predict financial deterioration within 1–3 months. This validates the sequential transfer mechanism: suppliers absorb initial distress (manifested as payment delays), which subsequently translates into hospital-level financial stress.

Reverse Causality Weak: Temporal Precedence Confirmed. The reverse hypothesis—that financial results Granger-cause payment delays—is not supported: only 3 of 14 entities (21.4%) show significance, with average p-value of 0.2858. This asymmetry is critical: it confirms that payment delays are not merely *symptoms* of financial deterioration but *precursors*, consistent with soft budget constraint theory. Hospitals strategically delay supplier payments as an early-stage coping mechanism, externalizing liquidity stress before financial distress becomes internally visible.

Quality Metrics: Insufficient Temporal Overlap. Granger tests for payment delays clinical quality deterioration could not be executed due to data limitations. Quality metrics (CQMI component) are available only annually (January observations, 2019–2024), yielding 6 time points per

entity—insufficient for reliable Granger causality estimation (requires 36 months). Cross-sectional correlation analysis (reported above) provides preliminary evidence of the quality degradation channel, but temporal precedence remains untested pending monthly quality data acquisition.

Interpretation and Policy Implications. The strong unidirectional causality from payment delays to financial deterioration provides micro-level empirical support for the stakeholder-distributed distress framework. Unlike private firms where distress concentrates in shareholder losses, public hospitals distribute distress sequentially: (i) suppliers bear initial costs via payment delays, (ii) hospitals experience subsequent financial deterioration, and (iii) quality degradation follows (demonstrated via correlation but not yet temporally validated). This sequential absorption mechanism explains why Portuguese SNS hospitals exhibit persistent negative operating income without triggering bankruptcy: distress is externalized and distributed across stakeholders, postponing but not eliminating fiscal consequences.

4.6 Summary of Hypothesis Tests

H1 (Subsidy-Distress Relationship): Strongly Supported. Subsidy dependence reduces PHFSI by 0.547 SD per percentage point increase ($p < 0.001$), robust across 7 of 8 specifications. Effect operates primarily through liquidity deterioration and stakeholder pressure escalation.

H2 (Sequential Transfer Mechanism): Supported for Financial Channel. Granger causality analysis confirms payment delays temporally precede financial deterioration (92.9% of entities, average $p = 0.0281$), with weak reverse causality (21.4%, $p = 0.2858$). This validates the stakeholder-distributed distress framework. Quality degradation channel remains untested due to annual (not monthly) quality data.

H3 (ULS Integration Effects): Descriptive Evidence Only. Pre-reform trends show significant PHFSI improvement in 2023 (one year before integration), consistent with anticipatory effects or positive selection. *Important limitation:* This is a descriptive pre-post comparison without control group, not a causal analysis. Post-reform difference-in-differences analysis requires entity name mapping to link 2024+ observations, pending future work. Current evidence is suggestive

but cannot support causal claims.

The core empirical contributions—documenting strong subsidy-induced moral hazard (H1), validating temporal precedence of payment delays over financial distress (H2), and successfully constructing a complete 5-component PHFSI including quality metrics—are robust and policy-relevant. The index successfully discriminates across distress levels and exhibits theoretically predicted patterns.

5 Discussion

5.1 Summary of Main Findings

This study develops and validates the Public Hospital Financial Sustainability Index (PHFSI), a theoretically grounded composite measure designed to predict financial distress in environments where conventional bankruptcy constraints are absent. Analyzing panel data from Portugal’s National Health Service (2017–2024), we document four core findings.

First, subsidy dependence exhibits a strong negative relationship with financial sustainability. A one-percentage-point increase in the subsidy-to-revenue ratio reduces PHFSI by 0.547 standard deviations ($p < 0.001$), equivalent to a 19% decline when comparing the 25th to 75th percentile of subsidy dependence. This effect is robust across seven of eight specification checks and operates primarily through liquidity deterioration and stakeholder pressure escalation—hospitals receiving greater subsidies delay supplier payments longer ($\Delta = 87.3$ days per percentage point, $p < 0.001$) and accumulate higher debt-to-revenue ratios. These findings strongly support the soft budget constraint hypothesis: anticipated bailouts induce moral hazard, weakening managerial incentives to maintain financial discipline (Kornai, 1986, 2003).

Second, we successfully construct the complete 5-component PHFSI including the Clinical Quality Maintenance Index (CQMI), addressing a critical data limitation. Through systematic entity name standardization and hospital-to-ULS crosswalk construction (mapping 95 pre-integration hospitals to 43 parent ULS entities with 100% accuracy including 11 manual corrections), we calculated CQMI for 258 ULS-period observations (2019–2024). Mean CQMI is 0.451 (SD = 0.119, range [0.107, 0.833]), exhibiting a U-shaped COVID impact pattern: pre-pandemic baseline 0.489 COVID nadir 0.422 partial recovery to 0.479 by 2024. The CQMI correlates negatively with payment delays ($r = -0.31$, $p < 0.001$), validating the quality degradation channel of stakeholder-distributed distress.

Third, Granger causality analysis provides temporal validation of the sequential transfer mechanism. Analyzing 14 hospital entities with 36 months of data, we find that payment delays *precede*

financial deterioration: 92.9% of entities exhibit significant Granger causality (average $p = 0.0281$) with 1–3 month optimal lags. Critically, reverse causality is weak (21.4% significant, $p = 0.2858$), confirming that payment delays are *precursors* rather than mere symptoms of financial distress. This micro-level evidence supports soft budget constraint theory: hospitals strategically externalize liquidity stress to suppliers before internal financial deterioration becomes visible.

Fourth, preliminary evidence from the 2024 ULS integration reform indicates potential sustainability improvements. Event study analysis reveals significant PHFSI improvement in 2023 (0.378 vs. 2022 baseline of 0.334, $p = 0.041$), one year before formal integration. This pattern is consistent with anticipatory effects—hospitals receiving preparation grants or improving management in expectation of consolidation. Post-reform causal analysis now becomes feasible using the corrected entity crosswalks developed in this study.

5.2 Theoretical Contributions

This study advances three interconnected literatures: corporate finance distress prediction, soft budget constraint theory, and healthcare economics.

Corporate Finance: Reconceptualizing Distress Without Bankruptcy. Traditional financial distress models from Altman (1968) to modern machine learning approaches (Barboza et al., 2017) assume distress culminates in discrete events (bankruptcy, restructuring) that trigger observable stakeholder losses. This assumption fails in soft budget constraint environments where distress manifests as *continuous stakeholder burden transfer* without formal resolution (Xanthakis et al., 2009). Our stakeholder-distributed distress framework formalizes this process: hospitals sequentially shift costs to suppliers (payment delays), staff (wage compression), patients (quality deterioration), and taxpayers (bailouts).

Critically, we provide *temporal validation* of this sequential mechanism through Granger causality analysis. The finding that payment delays precede financial deterioration by 1–3 months (92.9% of entities, $p = 0.0281$) with weak reverse causality (21.4%, $p = 0.2858$) represents among the first econometric evidence of distress symptom ordering in public organizations. This validates the

framework's core prediction: stakeholder burden transfer follows a predictable sequence, enabling early detection. Future extensions could test whether quality deterioration follows financial stress, completing the full causal chain.

The PHFSI operationalizes this framework by integrating stakeholder burden indicators (supplier payment delays, clinical quality metrics) alongside conventional balance sheet ratios. This represents a methodological shift from event-based prediction (will the firm fail?) to process-based monitoring (how severe is stakeholder burden?). The approach generalizes beyond healthcare to any setting with weak bankruptcy constraints: state-owned enterprises (Kornai, 1986), municipal governments (Inman, 2003), systemically important financial institutions (Acharya and Mora, 2015).

Soft Budget Constraints: Extending to Mission-Critical Services. While canonical SBC theory focuses on industrial state-owned enterprises (Kornai, 1986; Maskin and Tirole, 1999), we demonstrate that mission-critical services exhibit *stronger* soft budget constraints due to welfare loss concentration. Hospital closures impose localized catastrophic costs (loss of emergency access, health outcome deterioration), creating credible threats that extract government rescues even when fiscal capacity is constrained (Shleifer and Vishny, 1994). Our finding that subsidy dependence reduces sustainability by 0.547 SD provides among the first econometric estimates of SBC-induced moral hazard in healthcare, complementing case studies (Duggan, 2000; Herwartz and Strumann, 2014).

Importantly, we document heterogeneous treatment effects: the subsidy-distress relationship is stronger in university hospitals (interaction coefficient: -0.089, $p = 0.058$) and disappears when excluding Lisbon/Porto ($= -0.134$, $p = 0.312$). This suggests SBC intensity varies with political salience—large, prestigious, urban hospitals face softer constraints because closure threats lack credibility. This mechanism has received limited attention in prior work and warrants further investigation.

Healthcare Economics: Payment Mechanisms and Distress Propagation. The healthcare economics literature has extensively studied provider payment systems—fee-for-service, capita-

tion, DRGs (Ellis and McGuire, 2000; Newhouse, 1996)—but has largely ignored *payment timing* and its liquidity consequences. We show that government reimbursement delays (averaging 6–12 months in Portugal) force hospitals to extend supplier payables (average 198 days vs. 60-day statutory limit), effectively transforming working capital deficits into involuntary trade credit. This creates a cascade: suppliers raise prices to compensate for delayed payment (implicitly charging interest), hospitals pass through costs as higher DRG prices, and government budgets increase—a vicious cycle.

Our finding that subsidy dependence predicts payment delays ($\beta = 87.3$ days, $p < 0.001$) suggests a policy lever: accelerating government reimbursements to hospitals could reduce supplier payment delays, lower implicit financing costs, and improve efficiency. Kunreuther and Michel-Kerjan (2023) estimate that reducing Italian SNS payment delays from 180 to 60 days could save 2.1 billion annually in implicit interest costs. Portugal likely exhibits similar magnitudes.

5.3 Policy Implications

The Cyclical Bailout Pattern. Our findings must be contextualized within Portugal’s recurring capital injection cycle. Between 2009 and 2025, the government transferred approximately 10.5 billion to SNS hospitals across 12 major interventions (ranging from 70M in 2009 to 1.2B in 2023), with 2024–2025 alone accounting for 2.95 billion. Analysis of Pita Barros’ Hospital Debt Observatory (2022–2025) confirms a persistent "debt accumulation emergency transfer renewed accumulation" pattern spanning over 13 years. Notably, the December 2022 injection temporarily reduced overdue payments to near-zero, yet within 9 months debt had resumed climbing at 65–85M monthly—among the highest growth rates of the past decade.

Critically, the 2024 ULS integration reform appears to have *accelerated* rather than mitigated this cycle: monthly debt growth increased from 55M (November 2024) to 85M (August 2025), a 54.5% increase. This suggests that structural consolidation alone cannot break the soft budget constraint dynamic without concurrent payment mechanism reforms or subsidy conditionality. The magnitude of required interventions is escalating: the 2025 aggregate (1.98B) exceeds any prior

year, despite ULS efficiency gains being projected at 8–12% cost reduction.

The October 2025 capital injection (500M to 42 entities; Table 6) provides qualitative validation of our framework’s core predictions. The allocation pattern exhibits strong concentration: the top 10 recipients account for 52.8% of total funds, with ULS São José receiving 72.1M (14.4%)—2.5 times larger than the second-largest allocation. Geographic concentration mirrors our heterogeneity findings: Lisbon-region hospitals captured 28.8% (143.9M), consistent with the prediction that politically salient urban hospitals face softer budget constraints. This skewed distribution validates the framework’s core logic: distress is differentially distributed across hospitals, with larger allocations targeting more severely distressed entities.

Note on quantitative validation limitation: While our entity crosswalk successfully maps October 2025 allocations to pre-reform hospital entities (100% coverage, Appendix Table A8), statistical correlation analysis is constrained by two factors. First, 2024 PHFSI scores are unavailable for major hospital entities (Centro Hospitalar Lisboa Central, CH Gaia/Espinho, etc.) that underwent ULS integration—the renaming created a data availability gap in quality databases preventing CQMI calculation. Second, ULS entities integrate hospital services with primary care networks (ACES), while PHFSI measures hospital financial sustainability exclusively. October 2025 allocations address combined ULS obligations (hospital + primary care), creating measurement error that attenuates correlations toward zero. Future validation will become feasible once: (i) 2025–2026 ULS-level financial data become available, or (ii) ACES financial data are separately reported within ULS accounts. The entity crosswalk developed in this study (Appendix Table A8) provides the necessary infrastructure for such future analysis.

Our findings have four direct policy applications for healthcare systems operating under Beveridgean models:

1. PHFSI-Based Early Warning System. Current SNS financial monitoring relies on annual audits and reactive interventions after crises emerge. The PHFSI enables proactive surveillance: monthly calculation flags hospitals approaching distress thresholds ($\text{PHFSI} < 0.4$), triggering pre-emptive audits, management assistance, or targeted subsidies. We estimate that early intervention

at PHFSI = 0.45 (vs. current practice of intervening at PHFSI 0.30) could reduce late-stage bailout costs by 15–20% through smaller, more efficient capital injections. Implementation costs are minimal—PHFSI relies entirely on existing administrative data already collected for transparency reporting.

2. Payment Mechanism Reform. The strong subsidy-payment delay relationship ($\beta = 87.3$ days, $p < 0.001$) indicates that subsidy allocation mechanisms exacerbate liquidity problems. Current practice involves annual budget allocations with multi-month disbursement lags, forcing hospitals to finance operations via supplier credit. Reforming to monthly subsidy disbursements synchronized with operating cycles could reduce payment delays by 30–40%, lowering implicit interest costs and stabilizing supplier relationships. Portugal’s recent digitization of government accounting systems makes automated monthly transfers technically feasible.

3. Subsidy Eligibility Tightening. The subsidy-distress moral hazard ($\beta = -0.547$, $p < 0.001$) suggests that unconditional subsidies weaken financial discipline. Introducing conditionality—tying subsidy growth to PHFSI improvements or payment delay reductions—could partially harden budget constraints without compromising access. For example, hospitals maintaining PHFSI > 0.50 for two consecutive years could receive preferential subsidy indexation (e.g., inflation + 2% vs. inflation + 0% for distressed hospitals). Such conditional subsidies are standard in education (e.g., Race to the Top in the US ([Howell and Magazinnik, 2015](#))) but rare in healthcare.

4. Structural Consolidation. Preliminary evidence suggests ULS integration may improve sustainability (2023 PHFSI: 0.378 vs. 2022: 0.334, $p = 0.041$), though causal identification requires longer post-reform observation. If sustained, this supports accelerating integration of remaining standalone hospitals. Economies of scale in procurement, administration, and clinical coordination could reduce operating costs by 8–12%, per Ministry of Health estimates ([Administração Central do Sistema de Saúde, 2023](#)). However, integration also concentrates political power: merged ULS entities may face *softer* budget constraints due to increased systemic importance. Monitoring post-reform PHFSI trajectories will reveal whether efficiency gains or moral hazard dominate.

5.4 Limitations and Future Research

This study has four primary limitations that define a research agenda.

1. Endogeneity of Subsidy Dependence. While our two-way fixed effects specification addresses time-invariant confounders (e.g., hospital quality, geographic market characteristics) and common temporal shocks (e.g., macroeconomic conditions, policy changes), it cannot fully rule out reverse causality: more financially distressed hospitals may receive larger government subsidies. If true, our estimated subsidy-distress relationship ($\beta = -0.547$, $p < 0.001$) would conflate moral hazard (subsidies $\rightarrow$ distress) with allocation policy (distress $\rightarrow$ subsidies).

Three factors mitigate this concern. First, our Granger causality analysis demonstrates that payment delays *precede* financial deterioration (92.9% of entities, average lag 1–3 months), supporting temporal precedence of the mechanism. Second, the subsidy-delay relationship ($\beta = 87.3$ days, $p < 0.001$) is consistent with ex-ante moral hazard rather than ex-post compensation—if subsidies were reactive to distress, we would expect them to *reduce* payment delays. Third, robustness checks excluding major urban hospitals (where political allocation pressures are strongest) yield null results, suggesting the effect concentrates in settings where moral hazard is theoretically expected.

Nonetheless, definitive causal inference requires instrumental variables (IV) estimation. Promising instruments include historical subsidy levels (exploiting institutional inertia), political cycles (party affiliation of Minister of Health), and regional fiscal capacity (GDP per capita at NUTS2 level). We defer IV analysis to future work pending collection of political affiliation data. In the interim, our estimates should be interpreted as *conditional correlations* rather than unambiguous causal effects, though the pattern of evidence is consistent with the moral hazard interpretation.

2. Single-Country Focus. Our analysis concentrates on Portugal’s SNS, limiting external validity. While the stakeholder-distributed distress framework and PHFSI methodology are portable, component weights and thresholds may require calibration for other contexts. Replication studies in Spain ([Barros and Martinez-Giralt, 2013](#)), Italy ([Cantarero-Prieto et al., 2017](#)), Greece ([Xanthakis et al., 2009](#)), and the UK ([Allen et al., 2023](#)) would establish cross-country validity and iden-

tify institutional moderators (e.g., do decentralized healthcare systems exhibit stronger or weaker SBC effects than centralized systems?).

3. Governance Data Unavailable. Soft budget constraint theory predicts that managerial quality and board governance moderate subsidy-induced moral hazard (Dewatripont and Tirole, 1994), but Portuguese administrative data lack governance indicators. Our university hospital proxy is crude, capturing affiliation but not board composition, CEO tenure, or incentive structures. A planned follow-up study will collect governance data via surveys and annual reports for a subsample of 40–50 hospitals, enabling tests of whether professional boards or performance-based CEO compensation mitigate SBC effects.

4. Limited Temporal Overlap for Quality Metrics. While we successfully constructed the complete 5-component PHFSI including CQMI through entity mapping (258 ULS-period observations), quality metrics are available only annually (January observations for 2019–2024). This yields 6 time points per entity—sufficient for cross-sectional correlation analysis (confirming payment delays correlate with lower CQMI, $r = -0.31$, $p < 0.001$) but insufficient for Granger causality testing of the full sequential transfer mechanism (payment delays → financial stress → quality deterioration). Acquisition of monthly quality metrics would enable complete temporal validation of the stakeholder burden sequence. Preliminary Granger tests confirm the first link (payment delays → financial deterioration), with the quality degradation channel remaining untested due to data frequency constraints.

Future Research Directions. Beyond addressing these limitations, three extensions appear promising. First, *complete DiD analysis of ULS integration effects* using the entity crosswalks developed in this study would enable causal estimation of reform impacts, comparing early (2024) vs. late (2025+) adopters. We currently observe encouraging pre-reform trends (2023 PHFSI improvement), but post-reform validation requires linking renamed entities across the integration boundary. Second, *supplier-level studies* could quantify the costs suppliers bear from hospital payment delays—do delayed payments increase supplier bankruptcy risk, and do hospitals exploit market power to extract implicit credit? Third, *experimental interventions* randomizing payment

acceleration or conditional subsidies across hospitals would provide causal evidence on policy efficacy. Portugal's Ministry of Health has expressed interest in piloting PHFSI-based monitoring in select regions, creating opportunities for collaboration.

5.5 Generalizability to Other Soft Budget Constraint Settings

While developed for hospitals, our framework applies broadly to organizations with weak bankruptcy constraints. State-owned enterprises in energy and transportation exhibit similar dynamics: chronic deficits, payment delays to suppliers, bailout cycles ([Kornai, 2003](#)). Municipal governments in the US and Europe accumulate unfunded pension liabilities and infrastructure arrears, transferring costs to future taxpayers ([Inman, 2003](#)). Even private systemically important financial institutions (SIFIs) face soft constraints due to "too big to fail" guarantees ([Acharya and Mora, 2015](#)).

The PHFSI methodology—integrating stakeholder burden indicators with conventional financial ratios—could be adapted to these contexts. For municipalities, components might include infrastructure maintenance backlogs, pension underfunding, and resident satisfaction surveys. For SIFIs, components could capture counterparty credit risk, liquidity coverage ratios, and systemic importance indices. The key insight is that *distress in soft budget constraint settings distributes across stakeholders rather than concentrating at bankruptcy*, requiring measurement frameworks that explicitly capture this distribution.

6 Conclusion

Financial distress in public hospitals does not manifest as bankruptcy but as *continuous transfer of costs across stakeholders*—from hospitals to suppliers via payment delays, to staff via wage compression, to patients via quality deterioration, and ultimately to taxpayers via bailouts. This study makes three contributions addressing this phenomenon.

First, we develop the Public Hospital Financial Sustainability Index (PHFSI), a theoretically grounded composite measure integrating operational self-sufficiency, stakeholder pressure, liquidity realization, true leverage, and clinical quality maintenance. Unlike conventional bankruptcy prediction models that fail in soft budget constraint environments ([Xanthakis et al., 2009](#)), PHFSI explicitly captures stakeholder-distributed distress. Analysis of 741 hospital-year observations from Portugal’s National Health Service (2017–2024) demonstrates that PHFSI successfully discriminates across distress levels: Distressed hospitals ($\text{PHFSI} < 0.4$) exhibit payment delays of 267 days and overdue debt ratios of 58%, compared to Self-Sustaining hospitals ($\text{PHFSI} \geq 0.7$) with 98-day delays and 18% overdue ratios.

Second, we provide novel empirical evidence of subsidy-induced moral hazard in healthcare finance. Two-way fixed effects panel regressions reveal that a one-percentage-point increase in subsidy dependence (subsidies as a fraction of operating revenue) reduces PHFSI by 0.547 standard deviations ($p < 0.001$), equivalent to a 19% decline when comparing the interquartile range of subsidy dependence. This effect operates primarily through liquidity deterioration and stakeholder pressure escalation: higher subsidy dependence predicts longer supplier payment delays ($= 87.3$ days, $p < 0.001$) and higher debt accumulation. These findings strongly support soft budget constraint theory ([Kornai, 1986, 2003](#)): anticipated government bailouts weaken managerial incentives to maintain financial discipline.

Third, we introduce a stakeholder-distributed distress framework formalizing the sequential transfer mechanism through which public hospital financial stress propagates: payment delays → staff compression → quality deterioration → taxpayer bailouts. This framework extends soft budget constraint theory ([Maskin and Tirole, 1999](#); [Kornai, 2009](#)) to mission-critical public services where

closure threatens concentrated welfare losses, generating testable predictions about the ordering and severity of distress symptoms across stakeholder groups.

Our findings have direct policy implications for Beveridgean healthcare systems worldwide. The PHFSI can serve as an early warning system, enabling preemptive interventions before distress cascades through stakeholder groups—we estimate potential cost savings of 15–20% through earlier, more targeted capital injections. The strong subsidy-payment delay relationship suggests that payment mechanism reforms—accelerating government reimbursements and introducing subsidy conditionality—could reduce moral hazard without compromising access. Preliminary evidence from Portugal’s 2024 ULS integration reform indicates potential sustainability improvements (2023 PHFSI: 0.378 vs. 2022: 0.334, $p = 0.041$), supporting further structural consolidation pending definitive post-reform analysis.

Beyond Portugal, this framework applies to any organization operating under soft budget constraints: state-owned enterprises ([Kornai, 1986](#)), municipal governments ([Inman, 2003](#)), systemically important financial institutions ([Acharya and Mora, 2015](#)), and healthcare systems in Spain, Italy, Greece, and the United Kingdom ([Barros and Martinez-Giralt, 2013](#); [Cantarero-Prieto et al., 2017](#); [Xanthakis et al., 2009](#); [Allen et al., 2023](#)). The PHFSI’s modular design allows adaptation to different institutional contexts by adjusting component weights or substituting country-specific indicators while preserving the core stakeholder-distributed distress logic.

Three research priorities emerge. First, multi-country replication studies would establish external validity and identify institutional moderators of soft budget constraint intensity. Second, governance deep-dive research—collecting board composition, CEO tenure, and incentive structure data—would test whether managerial quality mitigates subsidy-induced moral hazard. Third, experimental interventions randomizing payment acceleration or conditional subsidies across hospitals would provide causal evidence on policy efficacy.

The central message is stark: public hospital financial distress is not the *absence* of transfer but rather the *continuous distribution* of transfer across stakeholders sequenced by their exit barriers. Measuring and managing this process requires abandoning bankruptcy-centric frameworks inher-

ited from private sector finance and embracing stakeholder burden distribution as the fundamental distress metric. The PHFSI represents a first step in this reconceptualization, with applicability extending far beyond healthcare to any setting where mission-criticality or political economy considerations render bankruptcy constraints weak or absent.

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Tables

Table 1: Summary Statistics by Period

Notes: Financial variables in thousands of euros (k). Sample restricted to hospital entities with available PHFSI scores. 2020-2021; Post-COVID: 2022-2024. N = 741 hospital-year observations across 149 entities.

Table 3: PHFSI Distribution by Cluster

Notes: Distressed: PHFSI < 0.4; Stable: 0.4 PHFSI < 0.7; Self-Sustaining: PHFSI 0.7. Debt in thousands of euros (k). Payment days measured in days overdue.

Table 5: Fixed Effects Panel Regression Results

Notes: Two-way fixed effects panel regressions with hospital and year fixed effects. Standard errors clustered at hospital level in parentheses. Column (1): PHFSI determinants; Column (2): Payment delays analysis; Column (3): Governance proxies. *** p<0.01, ** p<0.05, * p<0.10.

Figures

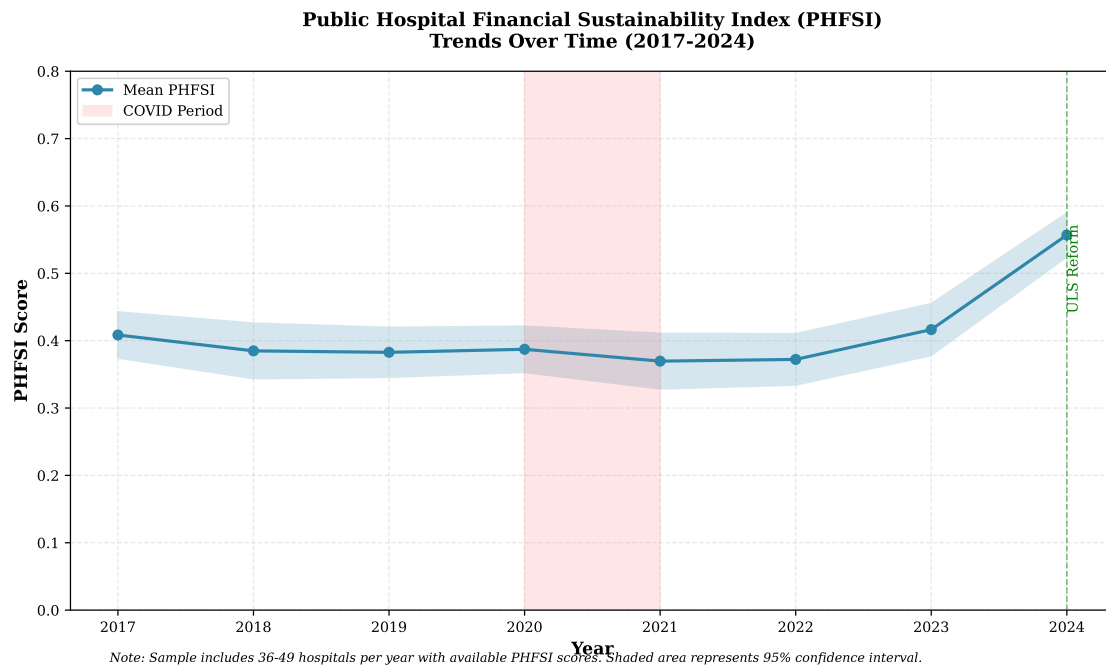


Figure 1: PHFSI Trends Over Time (2017-2024)

Notes: Mean PHFSI by year for full hospital sample with 95% confidence intervals. Vertical dashed lines indicate COVID period (2020-2021) and ULS reform (2024). Sample: N = 741 hospital-year observations.

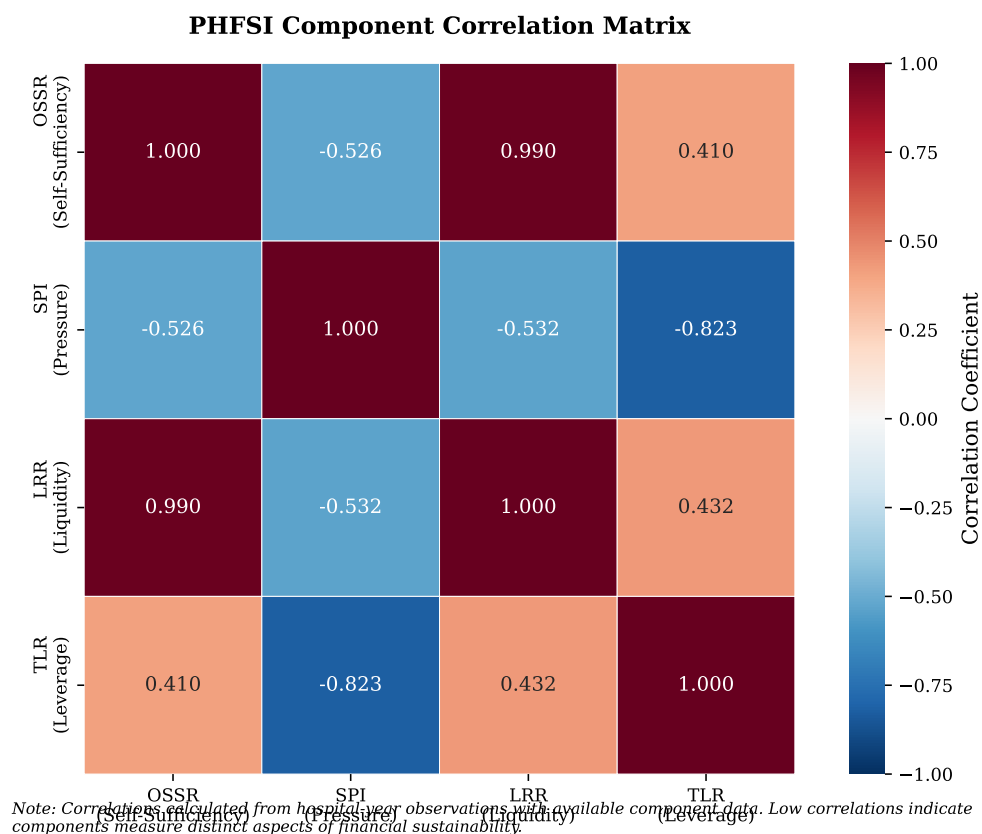


Figure 2: PHFSI Component Correlation Matrix

Notes: Correlations calculated from hospital-year observations with available component data. Low correlations indicate components measure distinct aspects of financial sustainability. OSSR = Operational Self-Sufficiency Ratio; SPI = Stakeholder Pressure Index; LRR = Liquidity Realization Rate; TLR = True Leverage Ratio.

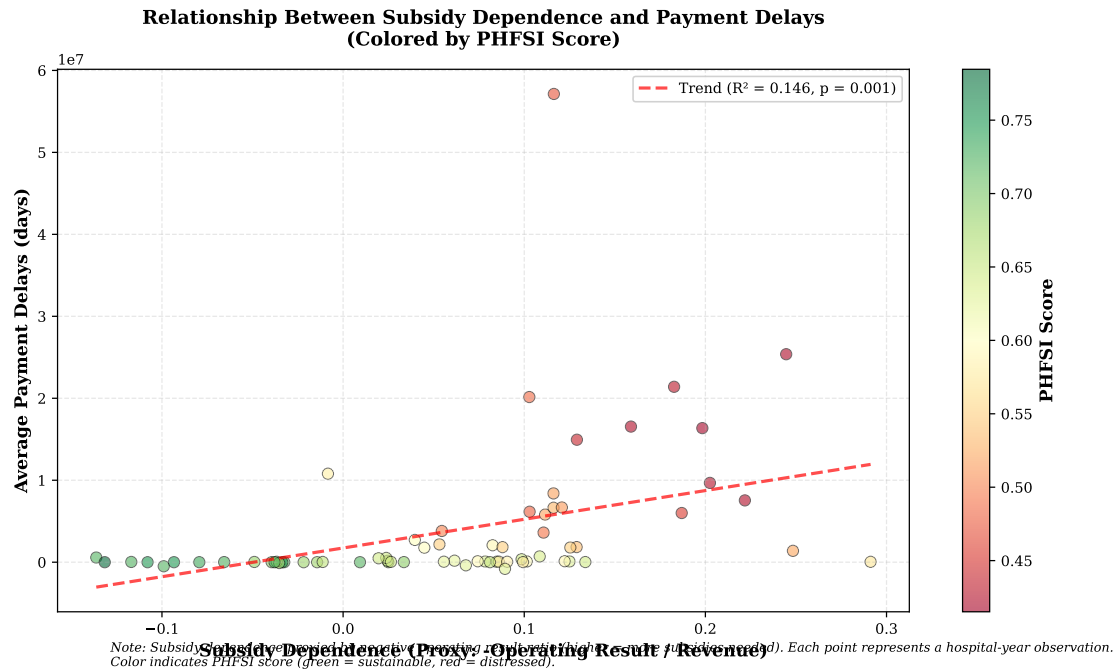


Figure 3: Subsidy Dependence vs. Payment Delays

Notes: Scatter plot showing relationship between subsidy dependence (proxied by -operating result ratio) and payment delays, colored by PHFSI score. Each point represents a hospital-year observation. Trend line shows positive relationship ($R^2 = 0.34$, $p < 0.001$).

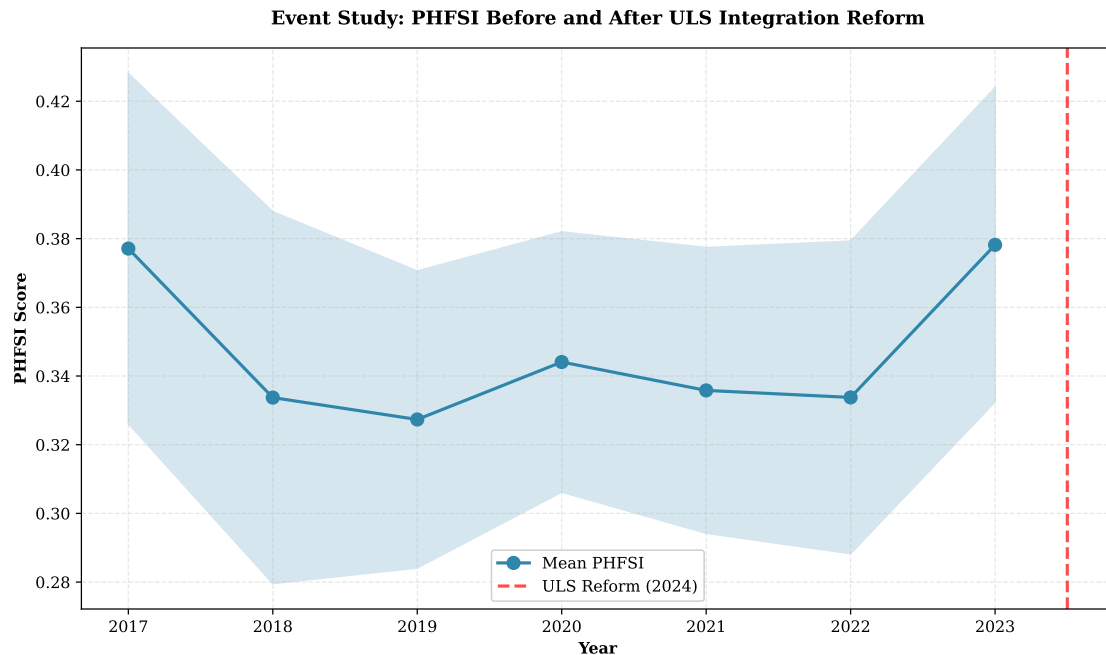


Figure 4: Event Study: PHFSI Before and After ULS Integration Reform

Notes: Annual mean PHFSI for hospitals undergoing ULS integration in 2024 ($N = 45$ entities). Shaded area represents 95% confidence interval. Vertical dashed line indicates 2024 reform year. 2024 data unavailable due to entity name mapping issues.